

Meet Agravat*Embedded Systems Engineer – Entry Level* meetagravat04@gmail.com |  +91 9327504499 |  [linkedin.com/in/agravat-meet](https://www.linkedin.com/in/agravat-meet)

PROFESSIONAL SUMMARY

Motivated B.Tech graduate in Electronics & Communication Engineering with hands-on experience in embedded firmware development, microcontroller interfacing, and real-time communication protocols (UART, I²C, SPI, CAN). Proven ability to design, prototype, and test hardware-software solutions that improve energy efficiency and system reliability. Seeking an entry-level role where I can apply strong C/C++ programming skills, knowledge of ARM/8051 architectures, and a solid grasp of the software development lifecycle to deliver robust embedded products.

CORE SKILLS

| **Programming** | C • C++ • Embedded C • Python |

| **Microcontrollers / Processors** | 8051 • ESP32 / ESP32-CAM • LPC2129 • ARM7 |

| **Communication Protocols** | UART • I²C • SPI • CAN (CANoe) |

| **Development Tools** | Keil µVision • Arduino IDE • GCC • VS Code • Proteus • JTAG • GDB |

| **Embedded Concepts** | Real-time firmware • State-machine design • ADC/PWM • Interrupt handling |

| **Project Management** | SDLC • Requirements analysis • Documentation • Team coordination |

EXPERIENCE**Embedded Systems Intern – Emertxe Information Technologies, Bangalore***Nov 2025 – Dec 2025*

- Developed firmware in **C** for a washing-machine simulation using PicsimLab, implementing a **state-machine control algorithm** that reduced cycle time by **≈15 %** in simulation.
- Applied full **SDLC** (requirements, design, coding, testing) to embed software, ensuring traceability and version control.
- Integrated and debugged peripheral interfaces (UART, I²C) on a development board, cutting hardware-setup time by **30 %** for subsequent team members.
- Produced technical documentation and test reports that met internal quality standards, facilitating smooth hand-off to senior engineers.

Professional Training – Vector India Pvt. Ltd.*Feb 2026 – Aug 2026*

- Completed an intensive **6-month program** covering embedded system development, automotive protocols, and industry-grade tools (CANoe, Keil µVision).
- Built and validated **CAN-bus communication** between two ECU prototypes, achieving **100 % message reliability** in lab tests.

- Programmed **ESP32-CAM** modules for IoT edge applications, integrating Wi-Fi and MQTT with secure firmware updates.
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PROJECTS

Smart Street Light System – Arduino

- Authored **Embedded C** code for LDR-based ambient light detection and PWM-controlled LED dimming, delivering **~40 % energy savings** versus constant-on lighting in simulation.
- Designed schematic and PCB layout in **Proteus**, and performed functional testing on an Arduino Uno prototype.

RFID-Based Attendance Management – ESP32

- Integrated **MFRC522 RFID** module over SPI and an I²C-connected 16×2 LCD, creating a real-time attendance logger for **20+ users**.
- Implemented secure tag authentication and data storage, reducing manual entry errors by **100 %**.

Temperature-Controlled Fan Speed Regulator – 8051

- Developed full firmware in **Embedded C** (Keil µVision) to read LM35 temperature via ADC, compute PWM duty cycle, and display values on a 7-segment display.
 - Demonstrated proportional fan speed control with **±2 °C** temperature accuracy in Proteus simulation.
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EDUCATION

Bachelor of Technology – Electronics & Communication Engineering

Government Engineering College, Bhavnagar (GTU) | 2022-2026

CGPA: **8.12 / 10**

Higher Secondary Certificate (Science – PCM)

Science School, Umralla | 2022 – 75 %

Secondary School Certificate

PMS School, Umralla | 2020 – 85 %

CERTIFICATIONS

- *Microcontroller Embedded C Programming: Absolute Beginners* – Udemy
 - *C Programming for Beginners – Master the C Language* – Udemy
 - *The Complete Embedded Systems Course* – Coursera (in progress)
 - *Introduction to the Internet of Things and Embedded Systems* – Coursera, UC Irvine
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ACHIEVEMENTS

- Represented college at **TechSprint Hackathon**, delivering an IoT-enabled prototype within a 24-hour window.

- Led technical setup for the **College Science & Technology Exhibition**, coordinating hardware demos for **5+** student teams.
 - Managed logistics for a college-level art competition, overseeing **50+** participants and ensuring on-time execution.
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TAILORING NOTES

1. **Keyword Alignment:** Integrated high-impact embedded-system terms (e.g., “firmware,” “real-time,” “CAN-bus,” “state-machine,” “SDLC”) throughout the resume to match typical ATS filters for embedded roles.
2. **Quantified Impact:** Added measurable results (e.g., “40 % energy savings,” “15 % cycle-time reduction,” “100 % message reliability”) to showcase concrete contributions.
3. **Reordered Content:** Prioritized the internship and professional training under Experience, placing the most relevant, employer-facing activities first, and re-phrased project bullet points to highlight technical depth and outcomes.